

Assignment Specification: Autonomous Agents in 3D Environments**Module:** Intelligent Systems (Joint Project with Graphics and Visualization)**Group Size:** Exactly 4 Students**Duration:** 4 Weeks**Total Weight:** 20% of the Total Grade**1. Project Overview**

Your group is tasked with designing and developing an interactive 3D or VR simulation/game. While the visual environment and models are assessed in the Graphics module, **this module's assessment focuses entirely on the AI, pathfinding logic, and dynamic search algorithms.**

2. Group Roles & Individual Responsibilities

To ensure equal contribution, your group of 4 must assign each member to one of the following specific cross-functional roles. **Your individual grade will be based on your Git commits for your specific IS responsibility.**

- **Student 1**
 - Custom Graph Formulation. Responsible for writing the scripts that programmatically extract the 3D world/NavMesh data into a mathematical graph (nodes and adjacency lists) for the search algorithms to use.
- **Student 2**
 - Dynamic Adaptation & Event Interception. Responsible for the logic that detects when a player alters the environment (e.g., drops a barricade) and dynamically severs the mathematical edges in the graph to trigger a recalculation.
- **Student 3**
 - A* Search & Heuristic Design. Responsible for programming the primary A* navigation algorithm, the priority queue, and mathematically justifying the heuristic function.
- **Student 4**
 - Secondary Search & Debug Visualizer. Responsible for implementing the secondary search algorithm (BFS/UCS) and programming the "Toggleable Debug Mode" that draws the search frontiers and calculated paths directly into the 3D engine view.

3. Project Milestones & Deliverables

- **Concept & Scope Pitch:** A live session presenting your concept, explaining how the 3D space will be abstracted into a graph, and declaring which student is taking which role.
- **Logic & Asset Prototype:** Search algorithms must be functional in the debug console/data structures (printing correct paths) before full 3D visual integration.
- **Final Showcase & Viva:** A live demonstration of the completed simulation, followed by individual technical questioning focusing on algorithmic complexity, heuristics, and graph theory.

4. Marking Rubric (100%)

Assessment Type	Specific Criterion	Max Marks	Expectations for Full Marks
GROUP (20%)	Overall, AI Cohesion	20	The individual algorithmic components (Graph, Searches, Recalculation, Debugger, Graphics Environment) communicate with each other seamlessly without crashing or throwing null reference errors.
	Concept & Scope	15	The project was well-scoped, mathematically sound, and the final product reflects a unified intelligent system that met the original collaborative vision.
INDIVIDUAL (80%)	Role-Specific Algorithm Accuracy	25	Evaluated strictly on the student's assigned role: • Flawless, programmatic extraction of 3D data into a highly efficient mathematical graph/adjacency list. • Real-time event interception reliably severs graph edges and triggers recalculation without infinite loops. • A* Search is flawlessly implemented, mathematically accurate, and utilizes an efficient priority queue. • Secondary search works perfectly, and the toggleable Debug Mode accurately draws the live mathematical data into the 3D space.
	Git Commits & Code Quality	15	Students show a consistent history of commits over the period for their specific logic. The code is clean, well commented, and modular.
	Algorithmic Complexity	25	Can accurately and mathematically explain the specific algorithms and the data structures they personally implemented.
TOTAL		100	